

You do not need to submit your answers to any of these questions.

Question 1. For each pair (a, b) given below, use Euclid's Algorithm to find $\gcd(a, b)$, and find $x, y \in \mathbb{Z}$ such that $\gcd(a, b) = xa + yb$.

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|-------------------|------------------|--------------------|
| (i) $(112, 12)$ | (ii) $(111, 11)$ | (iii) $(2024, 33)$ |
| (iv) $(126, 107)$ | (v) $(396, 455)$ | (vi) $(217, 308)$ |

Question 2. For each pair $(a, n) \in \{(11, 111), (217, 308), (126, 107)\}$, determine whether a has a multiplicative inverse in $\mathbb{Z}_n$, and find a^{-1} if it exists.

Question 3. Find the smallest positive integer x such that $x \equiv 2^{532} \pmod{31}$.

Question 4. Calculate $\varphi(n)$ for $n \in \{91, 56, 2024, 450, 15^{33}\}$, expressing your answer as a product of primes.

Question 5. Find $x \in \{0, 1, \dots, 13\}$ such that $x \equiv 5^{100000} \pmod{36}$.

Question 6. Find all values of $x \in \mathbb{Z}$ that satisfy

$$x^2 \equiv 23 \pmod{77}.$$

Question 7. Determine the plaintext corresponding to the following ciphertext obtained from a shift cipher cryptosystem:

fjnzcnjnyynol.

Question 8. The key for an affine cipher cryptosystem is $(a, b) = (3, 7)$.

- (i) Encrypt the plaintext message **eucalyptus**.
- (ii) Determine the plaintext corresponding to the ciphertext **AOHMBAPJ**.

Question 9. Given the usual identification between the English alphabet and $\mathbb{Z}_{26}$, determine the key for an affine cipher where **T** is the encryption of **k**, and **C** is the encryption of **p**.

Question 10. The key for a Vigenère cipher is **rotnnest**. Decrypt the ciphertext

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